

Final

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Table of contents

https://github.com/LeaLambdin/ENVS-193DS_spring-2026_final	2
Problem 1	4
a.	4
b.	4
c.	4
Problem 2	4
a.	4
b.	6
c.	6
d.	6
e.	7
f.	8
g.	8
h.	9
i.	11
j.	12
k.	13
Problem 3	13
a.	13
b.	14
c.	14
d.	16
e.	18
f.	18

GitHub Repository:

https://github.com/LeaLambdin/ENVS-193DS_spring-2026_final

```
#install packages  
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
v dplyr      1.2.1      v readr      2.2.0  
v forcats    1.0.1      v stringr    1.6.0  
v ggplot2     4.0.2      v tibble     3.3.1  
v lubridate  1.9.5      v tidyr      1.3.2  
v purrr       1.2.2  
-- Conflicts ----- tidyverse_conflicts() --  
x dplyr::filter() masks stats::filter()  
x dplyr::lag()     masks stats::lag()  
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(janitor)
```

Attaching package: 'janitor'

The following objects are masked from 'package:stats':

chisq.test, fisher.test

```
library(rstatix)
```

Attaching package: 'rstatix'

The following object is masked from 'package:janitor':

make_clean_names

The following object is masked from 'package:stats':

filter

```
library(DHARMA)
```

Registered S3 method overwritten by 'lme4':

```
method      from  
na.action.merMod car
```

This is DHARMA 0.5.0. For overview type '?DHARMA'. For recent changes, type `news(package = 'DHARMA')`.

Note that the default setting in `simulateResiduals` was changed to conditional simulations since version 0.5.0. To specify a different setting, use `simulateResiduals(simulation = "unconditional")` in `simulateResiduals()`. For more details, see `?simulateResiduals`.

```
#read in data  
nests <- read_csv("data/nest_data_final.csv")
```

Rows: 299 Columns: 10

-- Column specification -----

Delimiter: ","

chr (3): alive, bird_species, ant_species

dbl (7): height_cm, diameter_mm, case_control, case, canopy_width, canopy_le...

i Use ``spec()`` to retrieve the full column specification for this data.

i Specify the column types or set ``show_col_types = FALSE`` to quiet this message.

```
my_data <- read_csv("data/my_data.csv")
```

New names:

Rows: 33 Columns: 11

-- Column specification

----- Delimiter: "," chr

(5): Date, Time of Day, Sun exposure (min), Physical activity (min), UV ... dbl

(3): Sleep hours, Caffeine Intake (mg), Pages Read lgl (3): ...9, ...10, ...11

i Use ``spec()`` to retrieve the full column specification for this data. i

Specify the column types or set ``show_col_types = FALSE`` to quiet this message.

* `` -> `...9`

* `` -> `...10`

* `` -> `...11`

Problem 1

a.

In part 1, they likely used a logistic regression. The response variable is whether American Avocets used a microhabitat (yes/no), and the predictor variable is distance to the water edge. Logistic regression is commonly used when the response variable is binary and the goal is to determine whether a predictor affects the probability of an outcome.

In part 2, they likely used a chi-square test of independence. The variables are bird species (categorical) and habitat type (categorical), and the test evaluates whether there is an association between the two categorical variables.

b.

Part 1 rewrite:

American Avocet microhabitat use changed with distance from the water edge. Areas closer to or farther from the water edge were used differently by American Avocets.

Part 2 rewrite:

Habitat use differed among the three bird species. American Avocets, Forster's Terns, and Black-necked Stilts were associated with different habitat types within the San Francisco Bay.

c.

One additional variable that could influence American Avocet microhabitat use is native plant percent cover, measured as the percent of ground covered by native plants within a sampling plot. This is a continuous numeric variable measured in percent, and it could influence habitat use because plant cover may affect nesting protection, visibility, predator risk, and access to foraging areas.

Problem 2

a.

```

nests_clean <- nests |>

# keep variables needed for analysis
select(case_control, height_cm, ant_species) |>

# keep only the three focal ant species
filter(ant_species %in% c("AB", "RRB", "BBR")) |>

# reorder factor levels and add full species names
mutate(
  ant_species = factor(
    ant_species,
    levels = c("AB", "RRB", "BBR")
  ),
  ant_species_full = case_when(
    ant_species == "AB" ~ "Crematogaster sjostedti",
    ant_species == "RRB" ~ "Crematogaster mimosae",
    ant_species == "BBR" ~ "Crematogaster nigriceps"
  )
)

# display structure of cleaned data
str(nests_clean)

```

```

tibble [270 x 4] (S3: tbl_df/tbl/data.frame)
 $ case_control      : num [1:270] 1 0 0 0 0 1 0 0 0 1 ...
 $ height_cm        : num [1:270] 392 310 513 154 324 ...
 $ ant_species       : Factor w/ 3 levels "AB","RRB","BBR": 2 2 2 2 1 2 3 2 1 3 ...
 $ ant_species_full : chr [1:270] "Crematogaster mimosae" "Crematogaster mimosae" "Crematogaster mimosae" ...

```

```

# display 10 random rows
slice_sample(nests_clean, n = 10)

```

```

# A tibble: 10 x 4
  case_control height_cm ant_species ant_species_full
      <dbl>      <dbl> <fct>         <chr>
1         1        152. RRB          Crematogaster mimosae
2         1        212. RRB          Crematogaster mimosae
3         0        140. BBR          Crematogaster nigriceps
4         0        332. AB           Crematogaster sjostedti
5         0        134. RRB          Crematogaster mimosae

```

6	1	288.	RRB	<i>Crematogaster mimosae</i>
7	1	181	RRB	<i>Crematogaster mimosae</i>
8	0	112	BBR	<i>Crematogaster nigriceps</i>
9	0	103	RRB	<i>Crematogaster mimosae</i>
10	1	194	RRB	<i>Crematogaster mimosae</i>

b.

In this data set, a value of 1 indicates that a bird nest was present in the tree, while a value of 0 indicates that no bird nest was present. Biologically, these values represent whether a tree was used or not used as nesting habitat by birds.

c.

Tree height

Tree height is a continuous variable measured in centimeters. As tree height increases, the probability of nest occurrence may increase because taller trees provide more space and structure for nesting and may offer better protection from predators. This information was found in the Methods section (nest tree surveys/data collection) where tree characteristics, including height, were measured for nest and available trees.

Acacia ant species

Acacia ant species is a categorical variable with three levels in this analysis: *Crematogaster sjostedti*, *Crematogaster mimosae*, and *Crematogaster nigriceps*. Ant species may influence the probability of nest occurrence because different ant species vary in how aggressively they defend trees from browsing mammals, and birds preferentially nested in trees occupied by strongly defensive ant species. This information was discussed in the Introduction and study rationale, where the authors explain the relationship between ant mutualists and bird nest-site selection.

d.

```
# create a table describing the models that will be compared
model_table <- tibble(

  # model identifier
  model_number = c(1, 2),

  # response variable for each model
```

```

response_variable = c(
  "case_control",
  "case_control"
),

# predictors included in each model
predictors = c(
  "height_cm + ant_species",
  "height_cm + ant_species"
),

# indicate whether an interaction term is included
interaction_term = c(
  "No",
  "height_cm × ant_species"
)
)

# display model table
model_table

```

```

# A tibble: 2 x 4
  model_number response_variable predictors interaction_term
      <dbl> <chr>           <chr>           <chr>
1         1 case_control    height_cm + ant_species No
2         2 case_control    height_cm + ant_species height_cm × ant_species

```

e.

```

# run logistic regression model without interaction
# this model tests whether tree height and ant species predict nest occurrence
model_1 <- glm(
  case_control ~ height_cm + ant_species,
  data = nests_clean,
  family = binomial
)

# run logistic regression model with interaction
# this model tests whether the effect of tree height depends on ant species
model_2 <- glm(

```

```
case_control ~ height_cm * ant_species,
data = nests_clean,
family = binomial
)
```

f.

```
# compare AIC values for the two candidate models
AIC(model_1, model_2)
```

	df	AIC
model_1	4	258.7430
model_2	6	237.8042

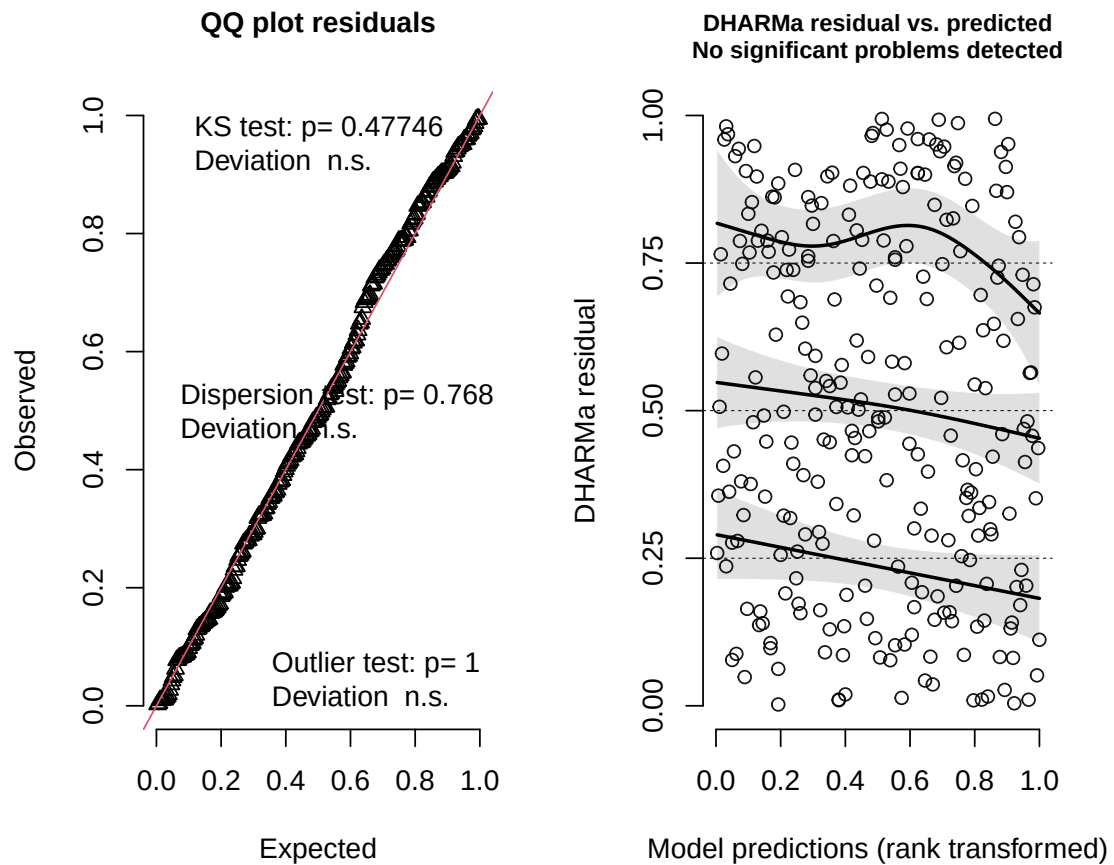
The best model as determined by Akaike's Information Criterion (AIC) included tree height, acacia ant species, and the interaction between tree height and acacia ant species as predictors of nest occurrence. This model had a lower AIC value than the model that included only the main effects of tree height and acacia ant species.

g.

```
# simulate residuals for the selected logistic regression model
# the selected model includes tree height, ant species, and their interaction
model_2_residuals <- simulateResiduals(
  fittedModel = model_2
)

# display diagnostic plots for the simulated residuals
plot(model_2_residuals)
```

DHARMA residual



h.

```
# create prediction data using the same ant species levels as the model
prediction_data <- expand_grid(
  height_cm = seq(
    min(nests_clean$height_cm),
    max(nests_clean$height_cm),
    length.out = 100
  ),
  ant_species = levels(nests_clean$ant_species)
) |>
mutate(
```

```

    ant_species = factor(
      ant_species,
      levels = levels(nests_clean$ant_species)
    ),
    ant_species_full = case_when(
      ant_species == "AB" ~ "Crematogaster sjostedti",
      ant_species == "RRB" ~ "Crematogaster mimosae",
      ant_species == "BBR" ~ "Crematogaster nigriceps"
    )
  )
)

# generate model predictions with standard errors
model_predictions <- predict(
  model_2,
  newdata = prediction_data,
  type = "link",
  se.fit = TRUE
)

# convert predictions from log-odds to probability scale
prediction_data <- prediction_data |>
  mutate(
    fit = plogis(model_predictions$fit),
    lower_ci = plogis(model_predictions$fit - 1.96 * model_predictions$se.fit),
    upper_ci = plogis(model_predictions$fit + 1.96 * model_predictions$se.fit)
  )

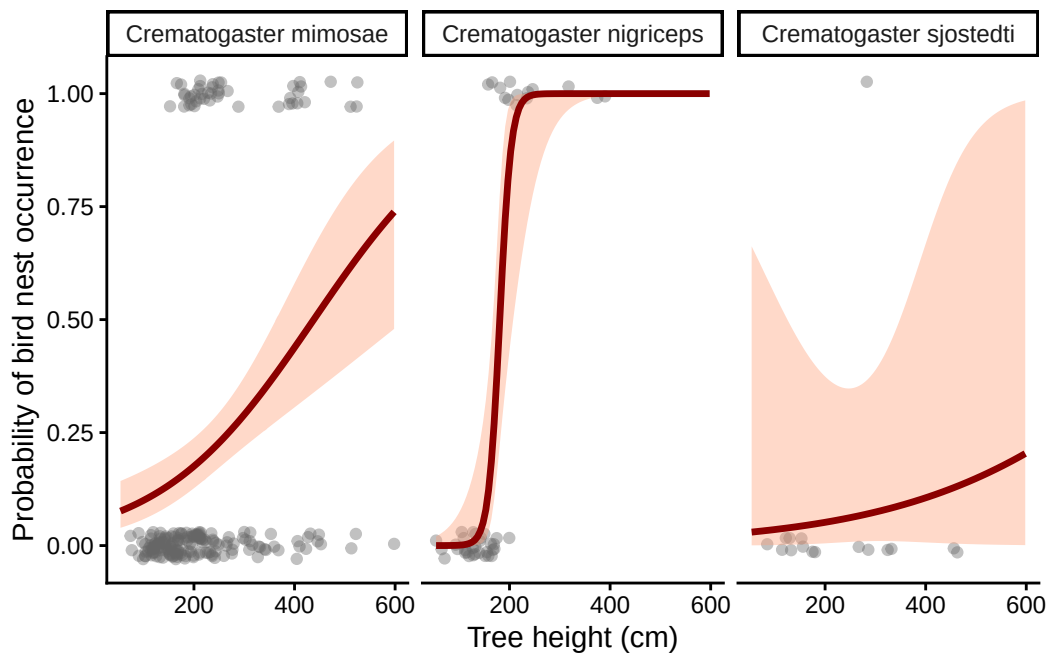
# plot observed data with model predictions and 95% confidence intervals
ggplot() +
  geom_jitter(
    data = nests_clean,
    aes(x = height_cm, y = case_control),
    width = 0,
    height = 0.03,
    alpha = 0.4,
    color = "gray40"
  ) +
  geom_ribbon(
    data = prediction_data,
    aes(x = height_cm, ymin = lower_ci, ymax = upper_ci),
    fill = "lightsalmon",
    alpha = 0.4
  )

```

```

) +
geom_line(
  data = prediction_data,
  aes(x = height_cm, y = fit),
  color = "darkred",
  linewidth = 1.2
) +
facet_wrap(~ ant_species_full) +
labs(
  x = "Tree height (cm)",
  y = "Probability of bird nest occurrence"
) +
theme_classic() +
theme(
  legend.position = "none"
)

```



i.

Figure 1. Probability of bird nest occurrence increases with tree height and varies among acacia ant species. Points represent individual trees, where values of 0 indicate no bird nest was present and values of 1 indicate a bird nest was present. Dark red lines represent predicted probabilities

of nest occurrence from the selected logistic regression model including tree height, acacia ant species, and their interaction. Shaded ribbons represent 95% confidence intervals around the model predictions. Panels show predictions for *Crematogaster mimosae*, *Crematogaster nigriceps*, and *Crematogaster sjostedti*. Data source: Lujan, E., Nielsen, R., Short, Z., Wicks, S., Nderitu Watetu, W., Malingati Khasoha, L., Palmer, T. M., Goheen, J. R., & Alston, J. M. (2023). Data, code, and metadata for: Symbiotic acacia ants drive nesting behavior by birds in an African savanna [Data set]. Zenodo. <https://doi.org/10.5281/zenodo.8373322>

j.

```
# create a new data frame with a tree height of 600 cm
# for each ant species included in the analysis
prediction_600 <- tibble(
  height_cm = 600,
  ant_species = factor(
    c("AB", "RRB", "BBR"),
    levels = levels(nests_clean$ant_species)
  )
)

# add full scientific names for clarity
prediction_600 <- prediction_600 |>
  mutate(
    ant_species_full = case_when(
      ant_species == "AB" ~ "Crematogaster sjostedti",
      ant_species == "RRB" ~ "Crematogaster mimosae",
      ant_species == "BBR" ~ "Crematogaster nigriceps"
    )
  )

# calculate predicted probabilities from the selected model
prediction_600 <- prediction_600 |>
  mutate(
    predicted_probability = predict(
      model_2,
      newdata = prediction_600,
      type = "response"
    )
  )
```

```
# display predicted probabilities
prediction_600
```

```
# A tibble: 3 x 4
  height_cm ant_species ant_species_full predicted_probability
  <dbl> <fct>      <chr>                                <dbl>
1     600 AB       Crematogaster sjostedti             0.205
2     600 RRB      Crematogaster mimosae              0.741
3     600 BBR      Crematogaster nigriceps             1
```

k.

As shown in Figure 1, the relationship between tree height and the probability of nest occurrence differed among acacia ant species. The probability of nest occurrence generally increased with tree height, but the strength of this relationship varied by ant species. At the tallest tree height in the dataset (600 cm), the predicted probability of nest occurrence was approximately 0.205 for *Crematogaster sjostedti*, 0.741 for *Crematogaster mimosae*, and 1.000 for *Crematogaster nigriceps*. These results suggest that both tree height and acacia ant species influence nest occurrence and that the effect of tree height depends on the ant species present. The authors suggest that birds preferentially nest in trees occupied by certain ant species because differences in ant behavior and tree defense can affect tree structure, browsing pressure from herbivores, and the suitability of trees as nesting habitat.

Problem 3

a.

How are the visualizations different from each other in the way you have represented your data?

The exploratory visualizations from Homework 2 used traditional statistical graphics, including a bar chart of average pages read across UV exposure categories and a scatterplot showing the relationship between sun exposure and pages read. In contrast, the affective visualization used flowers, petals, moons, colors, and flower size to represent multiple variables simultaneously, making the data more visually engaging and emphasizing the overall experience of each day rather than focusing on statistical summaries.

What similarities do you see between all your visualizations?

All of the visualizations used the same personal dataset and focused on relationships among UV exposure, sun exposure time, sleep, and pages read. Although the visual style differed,

each visualization was intended to communicate patterns in energy levels and how they may relate to sunlight exposure.

What patterns do you see in each visualization? Are these different between visualizations? If so, why? If not, why not?

The exploratory visualizations showed that there was no strong relationship between sun exposure and pages read, with a nearly flat trend line and similar average pages read across UV categories. The affective visualization showed the same overall pattern, but it made day-to-day differences in sleep, UV exposure, and sun exposure more noticeable because multiple variables were displayed together in a single flower. The patterns were generally similar because both visualizations used the same data, but the affective visualization emphasized individual daily experiences rather than statistical relationships.

What kinds of feedback did you get during week 9 in workshop or from the instructors? How did you implement or try those suggestions?

The feedback I received focused on making the visualization easier to interpret by clearly explaining what each visual element represented and improving the legend. I implemented this feedback by creating a detailed key that explained how flower color represented UV index, flower size represented sun exposure time, petals represented pages read, and moons represented sleep hours. These changes made the visualization easier for viewers to understand while preserving the creative design.

b.

One analysis I could use is a simple linear regression examining the relationship between sun exposure time (minutes per day) and pages read (pages per day). This analysis is appropriate because my research question asks whether increased sun exposure is associated with changes in energy level, which I measured as the number of pages read. The response variable, pages read, is a continuous quantitative variable, and the predictor variable, sun exposure time, is also a continuous quantitative variable measured in minutes per day. Linear regression is appropriate because it tests whether there is a relationship between two continuous variables and estimates how changes in sun exposure are associated with changes in pages read.

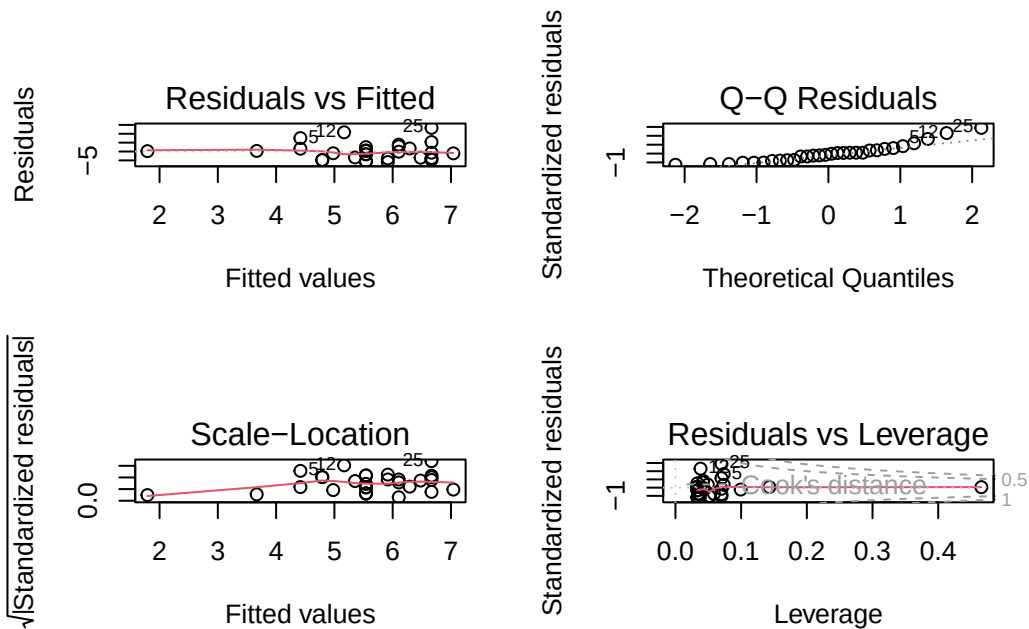
c.

```
# convert sun exposure from character to numeric
my_data <- my_data |>
  mutate(
    `Sun exposure (min)` = as.numeric(`Sun exposure (min)`)
  )
```

Warning: There was 1 warning in `mutate()`.
 i In argument: `Sun exposure (min) = as.numeric(`Sun exposure (min)`)`.
 Caused by warning:
 ! NAs introduced by coercion

```
# create a linear regression model relating pages read
# to sun exposure time
pages_model <- lm(
  `Pages Read` ~ `Sun exposure (min)`,
  data = my_data
)

# check model assumptions with diagnostic plots
par(mfrow = c(2, 2))
plot(pages_model)
```



```
par(mfrow = c(1, 1))
```

Assumption checks

Diagnostic plots were examined to assess the assumptions of linear regression. The Residuals vs Fitted plot did not show a strong pattern, suggesting the assumption of linearity was reasonably

met. The Q-Q plot showed some deviation from normality at the extremes, but most points followed the reference line. The Scale-Location and Residuals vs Leverage plots did not indicate major violations of homoscedasticity or influential observations. Overall, the assumptions were sufficiently met to proceed with the analysis.

Analysis

```
# display linear regression results
summary(pages_model)
```

Call:

```
lm(formula = `Pages Read` ~ `Sun exposure (min)`, data = my_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-5.9166	-3.6195	-0.3852	1.6771	13.3336

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	7.79114	2.04892	3.803	0.000712 ***
`Sun exposure (min)`	-0.03749	0.03124	-1.200	0.240152

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.782 on 28 degrees of freedom

(3 observations deleted due to missingness)

Multiple R-squared: 0.04892, Adjusted R-squared: 0.01495

F-statistic: 1.44 on 1 and 28 DF, p-value: 0.2402

Effect size

```
# calculate R-squared as an effect size
summary(pages_model)$r.squared
```

```
[1] 0.04892162
```

d.

```

# create visualization showing the linear regression model
# with underlying data and 95% confidence interval
ggplot(my_data, aes(x = `Sun exposure (min)`, y = `Pages Read`)) +

  # show individual observations
  geom_point(
    color = "darkgreen",
    size = 3,
    alpha = 0.7
  ) +

  # show model prediction line and confidence interval
  geom_smooth(
    method = "lm",
    se = TRUE,
    color = "darkorange",
    fill = "peachpuff"
  ) +

  # label axes with units
  labs(
    x = "Sun exposure time (minutes)",
    y = "Pages read (pages)",
    title = "Relationship between sun exposure time and pages read"
  ) +

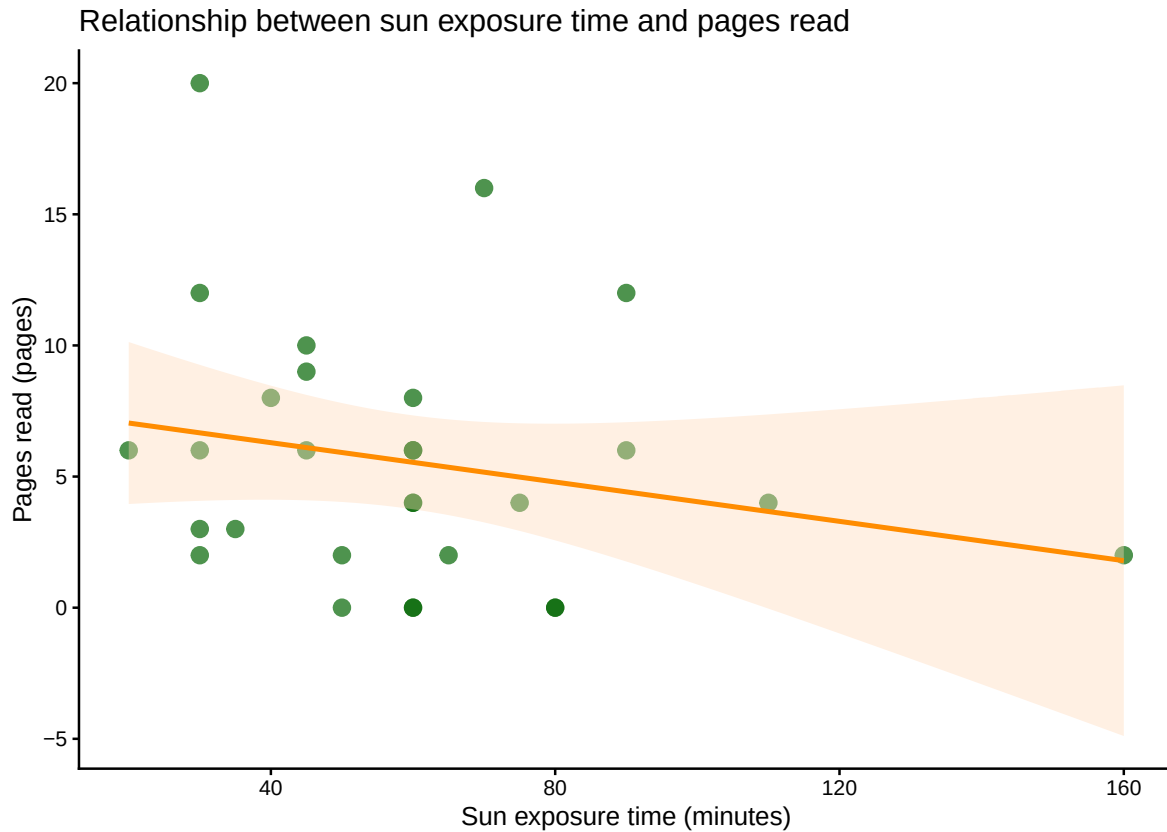
  # use non-default theme
  theme_classic()

```

`geom\_smooth()` using formula = 'y ~ x'

Warning: Removed 3 rows containing non-finite outside the scale range
(`stat\_smooth()`).

Warning: Removed 3 rows containing missing values or values outside the scale range
(`geom\_point()`).



e.

Figure 2. Relationship between sun exposure time and pages read. Dark green points represent individual daily observations of pages read and sun exposure time. The orange line represents the fitted linear regression model describing the relationship between sun exposure time and pages read. The light peach shaded band surrounding the regression line represents the 95% confidence interval around the model predictions. The figure shows the underlying data used in the analysis as well as the estimated trend between the two variables.

f.

There was a weak negative relationship between sun exposure time and pages read, with pages read tending to decrease slightly as sun exposure increased. However, this relationship was not statistically significant (linear regression: $\beta = -0.037$, $t = -1.20$, $p = 0.240$). Sun exposure time explained only a small proportion of the variation in pages read ($R^2 = 0.049$), suggesting that other factors were likely more important predictors of pages read in this dataset.